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MILITARY SPECIFICATION

SHAFTING, FLEXIBLE, AND CASING, STEEL (FOR REMOTE CONTROL)

This specification has been approved by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope.—This specification covers steel flexible-shaft and casing for applications requiring remote control.

1.2 Classification.—Casings shall be of the following finishes as specified (see 3.2.4, 6.2d, 6.3):

Zinc.
Cadmium.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications and standard, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

NN–B–591 —Boxes; Wood-Cleated-Fiberboard.
NN–B–601 —Boxes; Wood-Cleated-Plywood for Domestic Shipment.
NN–B–621 —Boxes; Wood, Nailed and Lock-Corner.
NN–B–631 —Boxes; Wood, Wirebound, for Domestic Shipment.
QQ–P–416 —Plating, Cadmium (Electrodeposited).
QQ–Z–325 —Zinc Plating (Electrodeposited).
LLL–B–631 —Boxes; Fiber, Corrugated (for Domestic Shipment).
LLL–B–636 —Boxes, Fiber, Solid (for Domestic Shipment).

MILITARY

JAN–P–105 —Packaging and Packing for Overseas Shipment — Boxes, Wood, Cleated, Plywood.
JAN–P–106 —Packaging and Packing for Overseas Shipment — Boxes, Wood, Nailed.
MIL–B–107 —Boxes, Wood, Wirebound (Overseas type).
JAN–P–108 —Packaging and Packing for Overseas Shipment — Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.
MIL–P–116 —Preservation, Methods of.
JAN–P–125 —Packaging and Packing for Overseas Shipment—Barrier-Materials, Waterproof, Flexible.
JAN–P–138 —Packaging and Packing for Overseas Shipment — Boxes, Wood, Fiberboard-Lined.
MIL–A–140 —Adhesive, Water-Resistant, Waterproof Barrier Material.
MIL–M–7911 —Marking, Identification of Aeronautical Equipment, Assemblies, and Parts.

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NAVY DEPARTMENT

General Specifications for Inspection of Material.

STANDARDS

MILITARY

MIL–STD–129—Marking of Shipments.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Drive-shaft.

3.1.1 *Construction.*—The drive-shaft shall consist of two or more successive layers of steel wire wound helically and in alternate pitch directions on a mandrel. Each successive layer shall be securely wound and free from kinks, loose turns, or other departures from best commercial practice. The ends shall be neatly and properly secured to prevent their unwinding. The shaft shall be suitable for either clock-wise or counter-clock-wise rotation.

TABLE I.—*Maximum permissible torsional angular deflection.*

Shaft diameter	Torque Load	Deflection in degrees per foot of shaft	
		Winding direction of outer layer	Unwinding direction of outer layer
Inches	*Pound-inches*		
0.130	1	10	11
.150	1	6	6
.188	3	10	10
.212	5	12	12
.250	5	11	12
.312	15	12	11
.375	30	6	7
.437	50	9	9

3.1.2 *Dimensions.*—The shaft dimensions shall be as specified (see 6.2). Unless otherwise specified, the tolerance on the diameter of the shaft shall be plus 0.000 minus 0.005 inches. The diameter at the secured ends of the shaft shall not exceed the nominal diameter of the shaft.

3.1.3 *Angular deflection.*—The maximum torsional angular deflection of shafts tested in compliance with 4.4.1 shall conform to table I.

3.1.4 *Permanent deformation.* — Shafts tested in compliance with 4.4.2 shall not be damaged or take a permanent set under the test loads specified in table II.

TABLE II.—*Torsional test loads.*

Shaft diameter	Torsional test load for direction indicated	
	Winding direction of outer layer	Unwinding direction of outer layer
Inches	*Pound-inches*	*Pound-inches*
0.130	7	5.5
.150	9	8.5
.188	12.5	12.5
.212	15	20
.250	22.5	27.5
.312	35	45
.375	80	87.5
.437	125	135

3.2 Casing.

3.2.1 *Construction.*—The casing shall be of steel and of such construction that when cut it will not be necessary to solder or braze the cut end to prevent unwinding. The joints shall be so constructed that when the casing is bent until the coils take up against each other on the inside of the loop, the outside will not separate to the extent that through openings exist between successive coils.

3.2.2 *Leakage.*—The flexible casing shall hold oil without showing any running leaks when tested in compliance with 4.4.3.

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3.2.3 *Dimensions.*—The inside and outside diameters of the casing shall be as specified (see 6.2 and 6.4).

3.2.4 *Finish.*—All steel parts of the casing shall be either zinc or cadmium plated in conformance with requirements for the type and class of QQ–Z–325 or QQ–P–416 specified in the contract order. If not otherwise specified in the contract or order, casings shall be zinc-coated, .0002 inches minimum thickness in conformance with type 1, class 3 of Specification QQ–Z–325 (see 6.3).

3.3 **Interchangeability.**—All parts having the same manufacturer's part number shall be interchangeable.

3.4 **Marking.**—Each flexible shaft and length of casing shall be marked with the manufacturer's name or symbol, and part numbers, except that Air Force purchases shall be marked in conformance with specification MIL–M–7911.

3.5 **Workmanship.** — The workmanship shall be first class in every respect.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Lot.**—A lot shall consist of flexible shafts and casings of one size, manufactured under the same conditions, and offered for delivery at one time.

4.2 **Sampling for lot acceptance tests.**

4.2.1 *Angular deflection.* — Three shafts from the first 100 or less and one shaft from each additional 100 of the lot shall be tested.

4.2.2 *Permanent deformation.*—All shafts of the lot shall be tested.

4.2.3 *Leakage.*—One specimen 12 inches in length from each 100 feet of casing in the lot shall be tested.

4.2.4 *Casing finish.*—One sample shall be tested from each lot.

4.3 **Inspection.**

4.3.1 *Place.*—Unless otherwise specified, inspection shall be at the point of delivery.

4.3.2 *Naval purchases, general inspection procedures.*—For naval purchases, the general inspection procedures shall be in accordance with General Specifications for the Inspection of Material.

4.3.3 Each shaft and length of casing shall be visually examined to determine compliance with the requirements of this specification with respect to materials, construction, dimensions, finish, marking, and workmanship. Each shaft which does not comply with requirements of this inspection shall be subject to rejection.

4.4 **Tests**

4.4.1 *Angular deflection.*—Secure one end of the shaft and apply the torque specified in table I to the other end in each direction of rotation. The lot shall be subject to rejection if any sample fails to comply with requirements.

4.4.2 *Permanent deformation.* — With the shaft assembled in the casing secure one end of the shaft and apply the torque specified in table II to the other end in each direction of rotation. Each shaft which does not comply with this test shall be subject to rejection.

4.4.3 *Leakage.* — Fill the casing samples with light machine oil (Navy Symbol No. 2110, S.A.E. No. 10) and inspect for running leaks after bending each specimen until the coils take up against each other on the inside of the loop. The lot shall be subject to rejection if any sample fails to comply with requirements.

4.4.4 *Casing finish.*—The sample shall be tested for plating as required by the applicable specification.

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4.5 Reinspection.—The contractor shall have the option of removing defective portions of the lot or reworking the lot and resubmitting the material for inspection.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.—Unless otherwise specified, all shafting shall be coated with a rust-preventive compound conforming to preservative P–3 of Specification MIL–P–116 and packaged in accordance with method 1A of specification MIL–P–116.

5.2 Packing

5.2.1 *For domestic shipment.*—The flexible-shafts and casings, preserved and packaged as described in 5.1, shall be packed in wood-cleated fiberboard boxes, wood-cleated-plywood boxes, nailed wood boxes, wirebound wood boxes, corrugated fiber boxes, or solid fiber boxes conforming to Specification NN–B–591, NN–B–601, NN–B–621, NN–B–631, LLL–B–631, or LLL–B–636, respectively. The weight of wood boxes and contents shall not exceed approximately 200 pounds except where the weight of an individual item exceeds this amount; and of fiberboard boxes and contents, approximately 90 pounds.

5.2.2 *For oversea shipment.*—The flexible-shafts and casings, preserved and packaged as described in 5.1, shall be packed in cleated-plywood boxes, nailed wood boxes, wirebound boxes, fiberboard boxes, or fiberboard-lined wood boxes, conforming to Specification JAN–P–105, JAN–P–106, MIL–B–107, JAN–P–108, or JAN–P–138, respectively. The weight of wood boxes and contents shall not exceed approximately 150 pounds except where the weight of an individual item exceeds this amount; and of fiberboard boxes and contents, approximately 70 pounds.

5.2.2.1 Unless otherwise specified, the contents shall be inclosed within a sealed waterproof bag, or its equivalent, made from material conforming to Specification JAN–P–125, for case liner material. The seams and closures shall be completely sealed with a continuous seam at least three-quarters of an inch wide of water-resistant adhesive conforming to Specification MIL–A–140.

5.3 Marking.—In addition to any special marking required by the contract or order, shipments shall be marked in conformance with Standard MIL–STD–129. Unless otherwise specified, the nomenclature shall be as follows:

Shafting, Flexible, and Casing, Steel (*Applicable application, e.g., Speedometer Drive or Tachometer Drive).
Steel (*Dimensional Data), (Manufacturer's Part No. _____*)
Specification No. _____*.

6. NOTES

6.1 Intended use.—Flexible-shafting as covered by this specification is intended for transmitting motion under low loads where it is impracticable to use rigid shafts and gears.

6.2 Ordering data.—Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Nominal shaft dimensions and tolerances (see 3.1.2).
(c) Inside and outside diameters of the casing (see 3.2.3).
(d) Finish, if other than zinc plating, type I, class 3 of specification QQ–Z–325 is required (see 3.2.4 and 6.3).
(e) Whether domestic or overseas packing and marking are required.

6.3 Finish.—The attention of purchasers is invited to the option offered in 3.2.4. Casings are generally available commercially with zinc plating .0002 inches thick. Other zinc and cadmium plating finishes are obtainable only by special processing and procurement may involve delay.

* Indicates that applicable data is to be entered by the contractor.

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6.4 Casing sizes.—Table III lists some galanized steel casing sizes available commerially. Two wire type casings are made of wo formed wires wound together. Four wire ype casings are made of four wires wound ogether. These wires are not as heavy as in wo wire construction, resulting in a pitch hat stiffens the casing transversely. Infornation as to other sizes and types of castngs which are commercially available may e obtained from manufacturers.

TABLE III.—*Casing sizes.*

Largest shaft diameter normally used	Casing diameters		Type
	Inside diameter, minimum	Outside diameter, maximum	
Inch	Inch	Inch	
.130	.147	.238	4-wire
.150	.169	.238	4-wire
.150	.170	.258	4-wire
.188	.208	.317	4-wire
.212	.258	.360	4-wire
.250	.320	.500	2-wire
.312	.410	.625	2-wire
.375	.437	.667	2-wire
.437	.562	.875	2-wire

6.5 Table IV is provided for information only. The minimum operating radius given in the second column is the radius of the smallest curve in which the particular shaft should be operated or bent.

TABLE IV.—*Minimum operating radius and torsional breaking load.*

Shaft diameter	Minimum operating radius	Torsional breaking load for straighten shafts	
		Winding direction of outer layer	Unwinding direction of outer layer
Inches	*Inches*	*Pound-inches*	*Pound-inches*
0.130	3¼	14	11
.150	3½	18	17
.188	4	25	25
.212	8½	30	40
.250	4¼	45	55
.312	4½	70	90
.375	5	160	175
.437	6	250	270

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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